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Hybrid drive systems for well stimulation operations

Abstract

In accordance with presently disclosed embodiments, a hybrid drive system that uses multiple sources of mechanical energy to drive a pump is provided. The hybrid drive system may include a first mover for generating first mechanical energy, a pump, a drivetrain for providing first mechanical energy from the first mover to the pump, and a second mover within the drivetrain to generate and provide second mechanical energy to the pump. The multiple sources of mechanical energy may provide flexibility with respect to system design and allow for alternative sources of fuel and energy to be used to drive pumping systems. This may reduce the total diesel fuel consumption necessary to perform a well stimulation operation as well as provide for configurations in which diesel engines may be excluded from the pumping process in favor of alternative energy sources that typically do not have sufficient torque capacity to power a pump.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation of U.S. patent application Ser. No. 18/420,556, filed Jan. 23, 2024, which is a Continuation of U.S. patent application Ser. No. 17/738,995, filed May 6, 2022, now U.S. Pat. No. 11,913,316, issued Feb. 27, 2024, which is a Continuation of U.S. patent application Ser. No. 16/321,155 filed Jan. 28, 2019, now U.S. Pat. No. 11,421,673, issued Aug. 23, 2022, which is a U.S. National Stage Application of International Application No. PCT/US2016/050196 filed Sep. 2, 2016, which are incorporated herein by reference in their entirety for all purposes.

TECHNICAL FIELD

(1) The present disclosure relates generally to treatment operations for hydrocarbon wells, and more particularly, to hybrid drive systems for well stimulation operations.

BACKGROUND

(2) Hydrocarbons, such as oil and gas, are commonly obtained from subterranean formations that may be located onshore or offshore. The development of subterranean operations and the processes involved in removing hydrocarbons from a subterranean formation are complex. Subterranean operations involve a number of different steps such as, for example, drilling a wellbore at a desired

well site, treating and stimulating the wellbore to optimize production of hydrocarbons, and performing the necessary steps to produce and process the hydrocarbons from the subterranean formation.

(3) Treating and stimulating a well bore can include, among other things, delivering various fluids (along with additives, proppants, gels, cement, etc.) to the wellbore under pressure and injecting those fluids into the wellbore. One example treatment and stimulation operation is a hydraulic fracturing operation in which the fluids are highly pressurized via pumping systems to create fractures in the subterranean formation. The pumping systems typically include high-pressure, reciprocating pumps driven through conventional transmissions by diesel engines, which are used due to their ability to provide high torque to the pumps. Over the course of a fracturing operation, however, the diesel engines may consume thousands of gallons of diesel fuel, which is expensive and can be difficult to supply in sufficient quantities in a well site.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure and its features and advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a diagram illustrating an example system for treatment operations, according to aspects of the present disclosure;

(3) FIG. 2 is a diagram illustrating another example system for treatment operations, according to aspects of the present disclosure; and

(4) FIG. 3 is a diagram illustrating an example pumping system, according to aspects of the present disclosure.

DETAILED DESCRIPTION

(5) Illustrative embodiments of the present disclosure are described in detail herein. In the interest of clarity, not all features of an actual implementation are described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation specific decisions must be made to achieve developers' specific goals, such as compliance with system related and business related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of the present disclosure. Furthermore, in no way should the following examples be read to limit, or define, the scope of the disclosure.

(6) The terms “couple” or “couples” as used herein are intended to mean either an indirect or a direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection, or through an indirect mechanical or electrical connection via other devices and connections. The term “fluidically coupled” or “in fluid communication” as used herein is intended to mean that there is either a direct or an indirect fluid flow path between two components.

(7) The present disclosure is directed to a hybrid drive system that uses multiple sources of mechanical energy to drive a pump. The multiple sources of mechanical energy may provide flexibility with respect to system design and allow for alternative sources of fuel and energy to be used to drive on-site pumping systems. This may reduce the total diesel fuel consumption necessary to perform a well stimulation operation as well as provide for configurations in which diesel engines may be entirely excluded from the pumping process in favor of alternative mechanical energy sources, such as spark-ignited natural gas engines, that typically do not have sufficient torque capacity to power a well stimulation pump. Additionally, the use of a second source of mechanical energy may increase the useful life of pumping systems by providing a second system that can account for the reduced output torque that is characteristic of aging engines and motors.

(8) FIG. 1 is a diagram illustrating an example system **100** for treatment operations, according to

aspects of the present disclosure. The system **100** includes a fluid management system **110** in fluid communication with a blender system **120**. The blender system **120** may in turn be in fluid communication with one or more pump systems **130** through a fluid manifold system **140**. The fluid manifold system **140** may provide fluid communication between the pump systems **130** and a wellbore **150**. In use, the fluid management system **110** may receive water or another fluid from a fluid source **115** (e.g., a ground water source, a pond, one or more frac tanks), mix one or more fluid additives into the received water or fluid to produce a treatment fluid with a desired fluid characteristic, and provide the produced treatment fluid to the blender system **120**. The blender system **120** may receive the produced treatment fluid from the fluid management system **110** and mix the produced treatment fluid with a proppant, such as sand, or another granular material **125** to produce a final treatment fluid that is directed to the fluid manifold **140**. The pump systems **130** may then pressurize the final treatment fluid to generate pressurized final treatment fluid that is directed into the wellbore **150**, where the pressurized final treatment fluid generates fractures within a formation in fluid communication with the wellbore **150**.

(9) An example one of the pump systems **130** may comprise a first mover **130a**, a pump **130b**, and a drive train **130c**. As used herein, a mover may comprise any device that converts energy into mechanical energy to drive a pump. Example movers include, but are not limited to, electric motors, hydrocarbon-driven or steam engines, turbines, etc. The drive train **130c** may be removably coupled to the first mover **130a** and the pumps **130b** through one or more drive shafts (not shown), and may comprise a transmission **130d** with one or more gears that transmits mechanical energy from the first mover to the pump **130b**. For instance, to the extent the pumps **130b** comprise reciprocating pumps, the mechanical energy may comprise torque that drives the pump **130b**.

(10) The drive train **130c** may further comprise a second mover **130e**. As depicted, the second mover **130e** is coupled to the transmission **130d** between the transmission **130d** and the pump **130b**. In the embodiment shown, the second mover **130e** may receive mechanical energy from the first mover **130a** through the transmission **130d** and provide the received mechanical energy to the pump **130b** augmented by mechanical energy generated by the second mover **130e**. It should be appreciated, however, that the orientation of the second mover **130e** with respect to the first mover **130a**, transmission **130d**, and the pump **130b** is not limited to the embodiment shown. In other embodiments, the second mover **130e** may be positioned between the transmission **130d** and the first mover **130a**, for instance, or between elements of the transmission **130d** itself. In yet other embodiments, the second mover **130e** may be incorporated into the transmission **130d** as part of a hybrid transmission system through which power from both the first mover **130a** and second mover **130e** are provided to the pump **130b**.

(11) The first mover **130a** and second mover **130e** may receive energy or fuel in one or more forms from sources at the wellsite. The energy or fuel may comprise, for instance, hydrocarbon-based fuel, electrical energy, hydraulic energy, thermal energy, etc. The sources of energy or fuel may comprise, for instance, on-site fuel tanks, mobile fuel tanks delivered to the site, electrical generators, hydraulic pumping systems, etc. The first mover **130a** and second mover **130e** may then convert the fuel or energy into mechanical energy that can be used to drive the associated pump **130b**.

(12) In the embodiment shown, the first mover **130a** may comprise an internal combustion engine such as a diesel or dual fuel (e.g., diesel and natural gas) engine and the second mover **130e** may comprise an electric motor. The internal combustion engine **130a** may receive a source of fuel from one or more fuel tanks (not shown) that may be located within the pumping system **130** and refilled as necessary using a mobile fuel truck driven on site. The electric motor **130e** may be electrically coupled to a source of electricity through a cable **130f**. Example sources of electricity include, but are not limited to, an on-site electrical generator, a public utility grid, one or more power storage elements, solar cells, wind turbines, other power sources, or one or more combinations of any of the previously listed sources.

(13) As depicted, the source of electricity coupled to the second mover **130e** comprises a generator **160** located at the well site. The generator may comprise, for instance, a gas-turbine generator or an

internal combustion engine that produces electricity to be consumed or stored on site. In the embodiment shown, the generator **160** may receive and utilize natural gas from the wellbore **150** or from another wellbore in the field (i.e., “wellhead gas”) to produce the electricity. As depicted, the system **100** may include gas conditioning systems **170** that may receive the gas from the wellbore **150** or another source and condition the gas for use in the generator **160**. Example gas conditioning systems include, but are not limited to, gas separators, gas dehydrators, gas filters, etc. In other embodiments, conditioned natural gas may be transported to the well site for use by the generator.

(14) The system **100** may further include one or more energy storage devices **180** that may receive energy generated by the generator **160** or other on-site energy sources and store in one or more forms for later use. For instance, the storage devices **180** may store the electrical energy from the generator **160** as electrical, chemical, or mechanical energy, or in any other suitable form. Example storage devices **180** include, but are not limited to, capacitor banks, batteries, flywheels, pressure tanks, etc. In certain embodiments, the energy storage devices **180** and generator **160** may be incorporated into a power grid located on site through which at least some of the fluid management system **110**, blender system **120**, pump systems **130**, and gas conditioning systems **170** may receive power.

(15) In use, the first mover **130a** and second mover **130e** may operate in parallel or in series to drive the pump **130b**, with the division of power between the movers being flexible depending on the application. For instance, in a multi-stage well stimulation operation, the formation may be fractured (or otherwise stimulated) in one or more “stages,” with each stage corresponding to a different location within the formation. Each “stage” may be accompanied by an “active” period during which the pumps are engaged and pressurized fluids are being pumped into the wellbore **150** to fracture the formation, and an “inactive” period during which the pumps are not engaged while other ancillary operations are taking place. The transition between the “inactive” and “active” periods may be characterized by a sharp increase in torque requirement.

(16) In an embodiment in which the first mover **130a** comprises a diesel engine and the second mover **130e** comprises an electric motor, both the diesel engine and electric motor may be engaged to provide the necessary power, with the percentage contribution of each depending on the period in which the system **100** is operating. For instance, during the “inactive” and “active” periods in which the torque requirements are relatively stable, the diesel engine, which operates more efficiently during low or near constant speed operations, may provide a higher percentage (or all) of the torque to the pump than the electric motor. In contrast, during transitions between “inactive” and “active” states, the electric motor may supplant the diesel engine as the primary source of torque to lighten the load on the diesel engine during these transient operations. In both cases, the electric motor reduces the torque required by the diesel engine, which reduces the amount of diesel fuel that must be consumed during the well stimulation operation. It should be noted that power sources could be used during continuous operation or intermittently as needed, including during transmission gear-shift events.

(17) In addition to reducing the amount of diesel fuel needed to perform a well stimulation operation, the use of a first mover and a second mover in a pump system described herein may provide flexibility with respect to the types of movers that may be used. For instance, natural gas engines, i.e., internal combustion engines that use natural gas as their only source of combustion, are typically not used in oil field environments due to their limited torque capacity. By including two movers within the pump system **130**, the torque capacity of the natural gas engine may be augmented to allow the use of a natural gas engine within the pump system **130**. For instance, in certain embodiments, the first mover **130a** may comprise a natural gas engine and the second mover **130e** may comprise an electric motor that operates in series or parallel with the natural gas engine to provide the necessary torque to power the pump **130b**.

(18) In certain embodiments, the pump systems **130** may be electrically coupled to a controller **190** that directs the operation of the first and second movers of the systems **130**. The controller **190** may comprise, for instance, an information handling system that sends one or more control signals to the pump systems **130** to control the speed/torque output of the first and second movers. As used herein an information handling system may comprise any system containing a processor and a memory

device coupled to the processor containing a set of instructions that, when executed by the processor, cause the processor to perform certain functions. The control signals may take whatever form is necessary to communicate with the associated mover. For instance, a control signal to an electric motor may comprise an electrical control signal to a variable frequency drive coupled to the electric motor, which may receive the control signal and alter the operation of the electric motor based on the control signal. In certain embodiments, the controller **190** may also be electrically coupled to other elements of the system, including the fluid management system **110**, blender system **120**, pump systems **130**, generator **160**, and gas conditioning systems **170** in order to monitor and/or control the operation of the entire system **100**. In other embodiments, some or all of the functionality associated with the controller **190** may be located on the individual elements of the system, e.g., each of the pump systems **130** may have individual controllers that direct the operation of the associated first and second movers.

(19) It should be appreciated that only one example configuration is illustrated in FIG. **1** and that other embodiments and configurations are possible, depending on the types of movers and energy or fuel. In certain embodiments, some or all of the pumping systems **130** may include the same configuration, including the same types of first and second movers. The configurations of the individual pumping systems **130** and of the pumping systems generally may depend, for instance, on the available fuel and energy sources at the well site. For example, if a source of natural gas is more readily available than diesel fuel at a particular well site, the pumping systems **130** may be configured to utilize natural gas as a source of fuel/energy for both the first and second movers, which could include the use of a dual fuel or natural gas driven engine as the first mover and an electric motor powered by a natural-gas driven generator as the second mover.

(20) In certain embodiments, excess energy generated by the pumping systems **130** or other elements within the system **100** may be used as an energy source for the first and/or second movers. The excess energy may be used instead of or in addition to any of the energy and fuel sources described above. FIG. **2** is a diagram illustrating another example system **200** for treatment operations in which the excess energy is utilized, according to aspects of the present disclosure. As can be seen, the system **200** comprises similar pumping systems **240** to those described above with respect to FIG. **1**. Notably, each of the pumps systems **240** may comprise a first mover **240a**, a pump **240b** and a drivetrain **240c** comprising a transmission **240d** and a second mover **240e**. The second movers of the pumping systems **240** may comprise electric motors that function similarly to the electric motor described above with respect to FIG. **1**.

(21) In the embodiment shown, the second movers of the pumping systems **240** may themselves comprise sources of energy for the system **200**. In particular, as can be seen, the second movers of the pumping systems **240** may be coupled to each other and to an energy storage device **280**. During inactive periods, or periods with lower torque requirement by the pumps, the first movers of the system **200** may generate excess energy, particularly when the first movers comprise diesel engines that are left idling during “inactive” periods. During those periods, some or all of the second movers may function as generators, receiving the excess energy from the first movers and converting that excess energy into another form of energy for immediate use by other ones of the second movers within the system or for storage within the energy storage device **280**. For instance, where the first movers comprise diesel engines and the second movers comprise electric motors, some or all of the electric motors may also function as electric generators used to generate electricity using the excess torque generated by the diesel engines, and that electricity may be consumed by other ones of the electric motors to immediately reduce the fuel consumption of the associated diesel engines and/or stored in the energy storage device **280** for later use.

(22) Similarly, where the first movers comprise diesel engines and the second movers comprise hydraulic motors driven by pressurized hydraulic fluids, some or all of the hydraulic motors may use excess torque generated by the diesel engines to pressurize the hydraulic fluids for use by other ones of the hydraulic motors within the system **200** and/or for storage within the energy storage device **280** in the form of pressurized tank of hydraulic fluid. Other configurations are possible within the scope

of this disclosure.

(23) The embodiment show in FIG. 2 could also be used to increase the load on the engines (e.g., first movers) when the system is operating in cold ambient temperatures. The increased load may help to raise the exhaust temperatures of the engines during cold weather. This may enable heat sensitive aftertreatment emission devices to operate more efficiently and reliably, with less clogging of those systems as experienced during light loading of the engine with low exhaust temperatures. The excess motive energy output from the pumping systems 240 during this cold weather operation of the pumps may be converted into another form of energy via the second movers for immediate use by one of the other second movers or for storage in the energy storage device 280 for later use.

(24) FIG. 3 illustrates an example pumping system 300, according to aspects of the present disclosure. The pumping system 300 may be used, for instance, as one or more of the pumping systems described above with reference to FIGS. 1 and 2. As depicted, the system 300 comprises a first mover 302 in the form of a diesel engine coupled to a reciprocating pump 304 through a hybrid transmission system 306 into which a second mover in the form of an electric motor (or electric motor/generator) is integrated. The first mover 302, pump 304, and transmission system 306 are mounted on a trailer 308 coupled to a truck 310. The truck 310 may comprise, for instance, a conventional engine that provides locomotion to the truck 310 and trailer 310 through a hybrid transmission incorporating an electric motor or hydraulic system. The system 300 may further comprise an electrical connection 312, such as a cable, between the hybrid transmission of the truck 310 and the second mover in the pump transmission system 306.

(25) In use, the truck 310 and trailer 308 with the pumping equipment mounted thereon may be driven to a well site at which a fracturing or other treatment operation will take place. In certain embodiments, the truck 310 and trailer 308 may be one of many similar trucks and trailers that are driven to the well site. Once at the site the pump 304 may be fluidically coupled to a wellbore (not shown), such as through a fluid manifold, to provide treatment fluid to the wellbore. The pump 304 may further be fluidically coupled to a source of treatment fluids to be pumped into the wellbore. When connected, the diesel engine may be started to provide a primary source of torque to the pump 304 through the pump transmission system 306. The electric motor in the pump transmission system 306 similar may be engaged to provide a supplemental source of torque to the pump 304. As depicted, the electric motor in the pump transmission system 306 may receive energy directly from the hybrid transmission of the truck 310, such that the truck itself operates as an electrical generator for the pumping operation. In addition to energy from the truck 310 and the electric motor in the pump transmission system 306, the pump may receive electricity from other energy sources on the site, including a dedicated electrical generator on site or other pumping systems located on the site.

(26) Embodiments disclosed herein include: A. An apparatus including a first mover for generating first mechanical energy, a pump, a drivetrain, and a second mover. The drivetrain provides the first mechanical energy from the first mover to the pump, and the second mover is disposed within the drivetrain to generate and provide second mechanical energy to the pump. B. A system including a first pump system including a first mover for generating first mechanical energy, a pump, a drivetrain for providing first mechanical energy from the first mover to the pump, and a second mover within the drivetrain to generate and provide second mechanical energy to the pump. The system also includes a fluid manifold providing fluid communication between the pump and a wellbore, and at least one of a fluid management system and a blender unit providing a source of treatment fluids to the pump. C. A method including generating first mechanical energy with a first mover mechanically coupled to a pump, generating second mechanical energy with a second mover mechanically coupled to the pump, and directing fluid from the pump to a wellbore using the first mechanical energy and the second mechanical energy.

(27) Each of the embodiments A, B, and C may have one or more of the following additional elements in combination: Element 1: wherein the first mover includes one of a diesel engine or a dual fuel engine and the second mover includes at least one of an electric motor and a hydraulic motor. Element 2: wherein the first mover includes a natural gas spark-ignited engine and the second mover includes

at least one of an electric motor and a hydraulic motor. Element 3: wherein the drivetrain includes a transmission and the second mover is coupled between the first mover and the transmission or between the transmission and the pump. Element 4: wherein the drivetrain includes a hybrid transmission into which the second mover is integrated. Element 5: wherein the pump includes a hybrid pump into which the second mover is integrated. Element 6: further including a trailer onto which the first mover, pump, drivetrain and second mover are mounted, and a truck coupled to the trailer, wherein the truck includes a diesel engine and a hybrid transmission with an integrated electric generator. Element 7: wherein the second mover is coupled to and receives energy from the integrated electric generator of the hybrid transmission.

(28) Element 8: further including an energy storage device to provide a source of energy to at least one of the first mover and the second mover to generate the respective first and second mechanical energy. Element 9: further including an electrical generator coupled to the energy storage device and at least one of the first mover and the second mover. Element 10: further including a gas conditioning system to receive natural gas from a wellbore and provide conditioned natural gas to the electrical generator from which the electrical generator generates electricity. Element 11: further including a second pump system with an other first mover, an other pump, an other drivetrain, and an other second mover. Element 12: wherein the second mover and the other second mover include electric motors electrically connected to share electrical energy. Element 13: wherein the electric motors are further electrically connected to an energy storage system for providing electricity to at least one of electric motors and storing energy generated by at least one electric generator.

(29) Element 14: further including receiving the first mechanical energy at the pump through a drivetrain coupled between the first mover and the pump. Element 15: wherein generating second mechanical energy with the second mover includes generating second mechanical energy within the drivetrain. Element 16: wherein the first mover includes at least one of a diesel engine, a dual fuel engine, and a spark-ignited natural gas engine. Element 17: wherein the second mover includes at least one of an electric motor and a hydraulic motor. Element 18: wherein generating second mechanical energy with the second mover includes receiving at least one of electricity and pressurized hydraulic fluid from an energy storage device coupled to the second mover.

(30) Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the following claims.

Claims

1. A system, comprising: a well stimulation pump; a first mover for generating a first mechanical energy from a first fuel or first energy; a second mover for generating a second mechanical energy from a second fuel or second energy, wherein the second fuel or second energy is a different type of fuel or energy from the first fuel or first energy; and a controller for directing operation of the first mover and the second mover, wherein both the first mechanical energy and the second mechanical energy are used to drive the well stimulation pump, and wherein the controller is configured to: determine whether the well stimulation pump is in an active period in which the well stimulation pump is engaged to pump fluid to a wellbore; determine whether the well stimulation pump is in an inactive period in which the well stimulation pump is not engaged to pump fluid to the wellbore; and energize the first mover, the second mover, or both depending on whether the well stimulation pump is in the active period or the inactive period.
2. The system of claim 1, wherein the controller is further configured to: determine whether the well stimulation pump is in a transition period between the active period and the inactive period; and energize the first mover, the second mover, or both depending on whether the well stimulation pump is in the transition period.
3. The system of claim 1, further comprising a drivetrain for outputting a combination of the first and second mechanical energies to the well stimulation pump.

4. The system of claim 1, wherein the second mover is powered by electricity produced using natural gas, wherein at least a portion of the natural gas is obtained from the wellbore on-site.
5. The system of claim 1, further comprising a gas conditioning system to produce conditioned natural gas, wherein the conditioned natural gas is used to power an electrical generator to generate electricity.
6. The system of claim 1, further comprising a trailer onto which the first mover, the well stimulation pump, and the second mover are mounted.
7. The system of claim 1, wherein the first mover and the second mover are configured to operate in parallel to power the well stimulation pump.
8. The system of claim 1, wherein the first mover comprises an internal combustion engine or a dual fuel engine, and wherein the second mover comprises an electric motor.
9. The system of claim 1, further comprising a fluid manifold for providing fluid communication between the well stimulation pump and the wellbore.
10. The system of claim 9, further comprising a blender unit for preparing a treatment fluid, wherein the treatment fluid is provided to the fluid manifold via the well stimulation pump.
11. The system of claim 3, wherein the drivetrain comprises a transmission, and wherein the second mover is coupled between the first mover and the transmission.
12. The system of claim 3, wherein the drivetrain comprises a transmission, and wherein the second mover is coupled between the transmission and the well stimulation pump.
13. The system of claim 1, further comprising a generator coupled to the second mover, wherein the generator is configured to produce electricity to power the second mover.
14. The system of claim 13, wherein the generator produces the electricity from natural gas received from the wellbore on-site.
15. The system of claim 13, further comprising a storage device configured to: receive energy generated from the generator; and store the received energy for later use.
16. The system of claim 1, wherein the second mover is coupled to an on-site generator, a utility grid, solar cells, wind turbines, a hybrid transmission of a truck, or one or more combinations thereof.
17. The system of claim 1, further comprising a variable frequency drive coupled to the second mover, wherein the variable frequency drive is configured to receive a control signal from the controller and alter operation of the second mover based on the received control signal.
18. The system of claim 1, further comprising: a generator configured to produce electricity from natural gas obtained from the wellbore on-site; a fluid manifold for providing fluid communication between the well stimulation pump and the wellbore; and a blender unit for preparing a treatment fluid, wherein the treatment fluid is provided to the fluid manifold via the well stimulation pump, wherein the electricity produced by the generator powers at least a portion of the second mover, the fluid manifold, and the blender unit.
19. A system, comprising: a well stimulation pump; a first mover for generating a first mechanical energy, wherein the first mover comprises an internal combustion engine or a dual fuel engine; a second mover for generating a second mechanical energy, wherein the second mover comprises an electric motor; a drivetrain for outputting a combination of the first mechanical energy and the second mechanical energy to the well stimulation pump; and a controller for directing operation of the first mover and the second mover, wherein both the first mechanical energy and the second mechanical energy are used to drive the well stimulation pump, and wherein the controller is configured to: determine whether the well stimulation pump is in an active period in which the well stimulation pump is engaged to pump fluid to a wellbore; determine whether the well stimulation pump is in an inactive period in which the well stimulation pump is not engaged to pump fluid to the wellbore; determine whether the well stimulation pump is in a transition period between the active period and the inactive period; and energize the first mover, the second mover, or both depending on whether the well stimulation pump is in one of the active period, the inactive period, or the transition period.
20. The system of claim 19, wherein the second mover is powered by electricity produced using natural gas, wherein at least a portion of the natural gas is obtained from the wellbore on-site.

